

Labsheet-3

1. Write a Java program to input and display 5 integers using a 1-D array
import java.util.Scanner;

```
public class ArrayExample {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int[] numbers = new int[5];  
  
        // Input 5 integers  
        System.out.println("Enter 5 integers:");  
        for (int i = 0; i < 5; i++) {  
            numbers[i] = sc.nextInt();  
        }  
  
        // Display the integers  
        System.out.println("The entered integers are:");  
        for (int i = 0; i < 5; i++) {  
            System.out.println(numbers[i]);  
        }  
  
        sc.close();  
    }  
}
```

OUTPUT:-

Enter 5 integers:

10

20

30

40

50

The entered integers are:

10

20

30

40

50

2. Create a Java program to input 10 integers and display them in reverse order.

```
import java.util.Scanner;
```

```
public class ReverseArray {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int[] numbers = new int[10];  
  
        // Input 10 integers  
        System.out.println("Enter 10 integers:");  
        for (int i = 0; i < 10; i++) {  
            numbers[i] = sc.nextInt();  
        }  
  
        // Display in reverse order  
        System.out.println("Integers in reverse order:");  
        for (int i = 9; i >= 0; i--) {  
            System.out.println(numbers[i]);  
        }  
  
        sc.close();  
    }  
}
```

OUTPUT:-

Enter 10 integers:

1

2
3
4
5
6
7
8
9
10

Integers in reverse order:

10
9
8
7
6
5
4
3
2
1

3. Write a Java program to find the sum of all elements in a 1-D array.

```
import java.util.Scanner;
```

```
public class ArraySum {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);
```

```

int[] numbers = new int[5];
int sum = 0;

// Input array elements
System.out.println("Enter 5 integers:");
for (int i = 0; i < 5; i++) {
    numbers[i] = sc.nextInt();
}

// Calculate sum
for (int i = 0; i < 5; i++) {
    sum = sum + numbers[i];
}

// Display sum
System.out.println("Sum of all elements = " + sum);

sc.close();
}
}

```

OUTPUT:-

Enter 5 integers:

10

20

30

40

50

Sum of all elements = 150

4. Create a Java program to find the maximum element in an array.

```
import java.util.Scanner;
```

```

public class MaximumElement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

```

```

        int[] numbers = new int[5];

```

```

        // Input array elements

```

```

        System.out.println("Enter 5 integers:");
        for (int i = 0; i < 5; i++) {
            numbers[i] = sc.nextInt();
        }

        // Assume first element is maximum
        int max = numbers[0];

        // Find maximum element
        for (int i = 1; i < 5; i++) {
            if (numbers[i] > max) {
                max = numbers[i];
            }
        }

        System.out.println("Maximum element = " + max);

        sc.close();
    }
}

```

OUTPUT:-

Enter 5 integers:

12

45

23

67

34

Maximum element = 67

5. Write a Java program to count even and odd numbers in an array.

```
import java.util.Scanner;
```

```

public class EvenOddCount {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

```

```
        int[] numbers = new int[5];
```

```
        int even = 0;
```

```
        int odd = 0;
```

```

// Input array elements
System.out.println("Enter 5 integers:");
for (int i = 0; i < 5; i++) {
    numbers[i] = sc.nextInt();
}

// Count even and odd numbers
for (int i = 0; i < 5; i++) {
    if (numbers[i] % 2 == 0) {
        even++;
    } else {
        odd++;
    }
}

System.out.println("Number of even numbers = " + even);
System.out.println("Number of odd numbers = " + odd);

sc.close();
}
}

```

OUTPUT:-

Enter 5 integers:

10

15

22

31

40

Number of even numbers = 3

Number of odd numbers = 2

6. Create a Java program to copy all elements from one array to another.

```
import java.util.Scanner;
```

```

public class CopyArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] arr1 = new int[5];
        int[] arr2 = new int[5];

        // Input elements into first array
        System.out.println("Enter 5 integers:");
        for (int i = 0; i < 5; i++) {
            arr1[i] = sc.nextInt();
        }

        // Copy elements from arr1 to arr2
        for (int i = 0; i < 5; i++) {
            arr2[i] = arr1[i];
        }

        // Display copied elements
        System.out.println("Elements of the second array:");
        for (int i = 0; i < 5; i++) {
            System.out.println(arr2[i]);
        }

        sc.close();
    }
}

```

OUTPUT:-

Enter 5 integers:

10
20
30
40
50

Elements of the second array:

10
20

30

40

50

7. Write a Java program to input and display elements of a 3×3 matrix.

```
import java.util.Scanner;
```

```
public class Matrix3x3 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][] matrix = new int[3][3];

        // Input matrix elements
        System.out.println("Enter 9 elements of the matrix:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                matrix[i][j] = sc.nextInt();
            }
        }

        // Display matrix
        System.out.println("The 3 x 3 matrix is:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(matrix[i][j] + "\t");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

OUTPUT:-

Enter 9 elements of the matrix:

1

2

3
4
5
6
7
8
9

The 3 x 3 matrix is:

1	2	3
4	5	6
7	8	9

8. Create a Java program to find the sum of all elements in a matrix.

```
import java.util.Scanner;
```

```
public class MatrixSum {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int[][] matrix = new int[3][3];  
        int sum = 0;  
  
        // Input matrix elements  
        System.out.println("Enter 9 elements of the matrix:");  
  
        for (int i = 0; i < 3; i++) {  
            for (int j = 0; j < 3; j++) {  
                matrix[i][j] = sc.nextInt();  
                sum = sum + matrix[i][j];  
            }  
        }  
  
        // Display matrix  
        System.out.println("The matrix is:");  
  
        for (int i = 0; i < 3; i++) {  
            for (int j = 0; j < 3; j++) {  
                System.out.print(matrix[i][j] + "\t");  
            }  
        }  
    }  
}
```

```

        }
        System.out.println();
    }

    // Display sum
    System.out.println("Sum of all elements = " + sum);

    sc.close();
}
}

```

OUTPUT:-

Enter 9 elements of the matrix:

1

2

3

4

5

6

7

8

9

The matrix is:

1	2	3
---	---	---

4	5	6
---	---	---

7	8	9
---	---	---

Sum of all elements = 45

9. Write a Java program to find the sum of each row in a matrix.

```
import java.util.Scanner;
```

```
public class RowSum {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int[][] matrix = new int[3][3];  
  
        // Input matrix elements  
        System.out.println("Enter 9 elements of the matrix:");  
  
        for (int i = 0; i < 3; i++) {  
            for (int j = 0; j < 3; j++) {  
                matrix[i][j] = sc.nextInt();  
            }  
        }  
  
        // Find sum of each row  
        for (int i = 0; i < 3; i++) {  
            int sum = 0;  
  
            for (int j = 0; j < 3; j++) {  
                sum = sum + matrix[i][j];  
            }  
  
            System.out.println("Sum of row " + (i + 1) + " = " + sum);  
        }  
    }  
}
```

```
        sc.close();
    }
}
```

OUTPUT:-

Enter 9 elements of the matrix:

1

2

3

4

5

6

7

8

9

Sum of row 1 = 6

Sum of row 2 = 15

Sum of row 3 = 24

10. Create a Java program to find the sum of each column in a matrix.

```
import java.util.Scanner;
```

```
public class ColumnSum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][] matrix = new int[3][3];
```

```
// Input matrix elements

System.out.println("Enter 9 elements of the matrix:");

for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        matrix[i][j] = sc.nextInt();
    }
}

// Find sum of each column

for (int j = 0; j < 3; j++) {
    int sum = 0;

    for (int i = 0; i < 3; i++) {
        sum = sum + matrix[i][j];
    }

    System.out.println("Sum of column " + (j + 1) + " = " + sum);
}

sc.close();
}
```

OUTPUT:-

Enter 9 elements of the matrix:

1

2

3

4

5

6

7

8

9

Sum of column 1 = 12

Sum of column 2 = 15

Sum of column 3 = 18

11. Write a Java program to transpose a matrix.

```
import java.util.Scanner;
```

```
public class MatrixTranspose {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int[][] matrix = new int[3][3];
```

```
        // Input matrix elements
```

```
        System.out.println("Enter 9 elements of the matrix:");
```

```
        for (int i = 0; i < 3; i++) {
```

```
            for (int j = 0; j < 3; j++) {
```

```
                matrix[i][j] = sc.nextInt();
```

```
    }  
}  
  
// Display original matrix  
System.out.println("Original Matrix:");  
  
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j < 3; j++) {  
        System.out.print(matrix[i][j] + "\t");  
    }  
    System.out.println();  
}  
  
// Display transpose  
System.out.println("Transpose of Matrix:");  
  
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j < 3; j++) {  
        System.out.print(matrix[j][i] + "\t");  
    }  
    System.out.println();  
}  
  
sc.close();  
}  
}
```

OUTPUT:-

Enter 9 elements of the matrix:

1

2

3

4

5

6

7

8

9

Original Matrix:

1	2	3
---	---	---

4	5	6
---	---	---

7	8	9
---	---	---

Transpose of Matrix:

1	4	7
---	---	---

2	5	8
---	---	---

3	6	9
---	---	---

12. Create a Java program to add two matrices of the same size.

```
import java.util.Scanner;
```

```
public class MatrixAddition {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
int[][] matrix1 = new int[3][3];
int[][] matrix2 = new int[3][3];
int[][] sum = new int[3][3];

// Input first matrix
System.out.println("Enter elements of first matrix:");
for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        matrix1[i][j] = sc.nextInt();
    }
}

// Input second matrix
System.out.println("Enter elements of second matrix:");
for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        matrix2[i][j] = sc.nextInt();
    }
}

// Add two matrices
for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        sum[i][j] = matrix1[i][j] + matrix2[i][j];
    }
}
```

```
// Display result
System.out.println("Sum of the two matrices:");

for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        System.out.print(sum[i][j] + "\t");
    }
    System.out.println();
}

sc.close();
}
```

OUTPUT:-

Enter elements of first matrix:

1
2
3
4
5
6
7
8
9

Enter elements of second matrix:

9

8

7

6

5

4

3

2

1

Sum of the two matrices:

10 10 10

10 10 10

10 10 10

13. Write a Java program to multiply two matrices (basic case with 3×3 matrices).

```
import java.util.Scanner;
```

```
public class MatrixMultiplication {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int[][] matrix1 = new int[3][3];
```

```
        int[][] matrix2 = new int[3][3];
```

```
        int[][] result = new int[3][3];
```

```
        // Input first matrix
```

```
System.out.println("Enter elements of first matrix:");

for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        matrix1[i][j] = sc.nextInt();
    }
}

// Input second matrix

System.out.println("Enter elements of second matrix:");

for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        matrix2[i][j] = sc.nextInt();
    }
}

// Multiply the matrices

for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        for (int k = 0; k < 3; k++) {
            result[i][j] += matrix1[i][k] * matrix2[k][j];
        }
    }
}

// Display result

System.out.println("Result of matrix multiplication:");
```

```
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j < 3; j++) {  
        System.out.print(result[i][j] + "\t");  
    }  
    System.out.println();  
}
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter elements of first matrix:

1

2

3

4

5

6

7

8

9

Enter elements of second matrix:

9

8

7

6

5

4

3

2

1

Result of matrix multiplication:

30 24 18

84 69 54

138 114 90

14. Write a Java program to input and display elements of a $2 \times 2 \times 2$ 3-D array.

```
import java.util.Scanner;
```

```
public class ThreeDArray{
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int[][][] arr = new int[2][2][2];
```

```
        // Input elements
```

```
        System.out.println("Enter 8 elements:");
```

```
        for (int i = 0; i < 2; i++) {
```

```
            for (int j = 0; j < 2; j++) {
```

```
                for (int k = 0; k < 2; k++) {
```

```
                    arr[i][j][k] = sc.nextInt();
```

```

        }
    }
}

// Display elements

System.out.println("Elements of the 3-D array:");

for (int i = 0; i < 2; i++) {
    System.out.println("Block " + (i + 1) + ":");

    for (int j = 0; j < 2; j++) {
        for (int k = 0; k < 2; k++) {
            System.out.print(arr[i][j][k] + "\t");
        }
        System.out.println();
    }
}

sc.close();
}
}

```

OUTPUT:-

Enter 8 elements:

1

2

3

4

5

6

7

8

Elements of the 3-D array:

Block 1:

1 2

3 4

Block 2:

5 6

7 8

15. Create a Java program to find the sum of all elements in a 3-D array.

```
import java.util.Scanner;
```

```
public class ThreeDArraySum {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int[][][] arr = new int[2][2][2];  
        int sum = 0;  
  
        // Input elements  
        System.out.println("Enter 8 elements:");  
  
        for (int i = 0; i < 2; i++) {
```

```

        for (int j = 0; j < 2; j++) {
            for (int k = 0; k < 2; k++) {
                arr[i][j][k] = sc.nextInt();
                sum = sum + arr[i][j][k];
            }
        }
    }

    // Display sum
    System.out.println("Sum of all elements = " + sum);

    sc.close();
}
}

```

OUTPUT:-

Enter 8 elements:

1

2

3

4

5

6

7

8

Sum of all elements = 36

16. Write a Java program to count positive and negative numbers in a 3-D array.

```
import java.util.Scanner;

public class PositiveNegative3D {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int[][][] arr = new int[2][2][2];

        int positive = 0;

        int negative = 0;

        // Input elements

        System.out.println("Enter 8 elements:");

        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                for (int k = 0; k < 2; k++) {
                    arr[i][j][k] = sc.nextInt();

                    if (arr[i][j][k] > 0) {
                        positive++;
                    } else if (arr[i][j][k] < 0) {
                        negative++;
                    }
                }
            }
        }
    }
}
```

```
// Display results

System.out.println("Number of positive numbers = " + positive);

System.out.println("Number of negative numbers = " + negative);


    sc.close();
}
}
```

OUTPUT:-

Enter 8 elements:

10

-5

20

-8

15

-2

7

-3

Number of positive numbers = 4

Number of negative numbers = 4

17. Create a Java program to find the maximum element in a 3-D array.

```
import java.util.Scanner;
```

```
public class Maximum3DArray {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

```
int[][][] arr = new int[2][2][2];
```

```
// Input elements
```

```
System.out.println("Enter 8 elements:");
```

```
for (int i = 0; i < 2; i++) {  
    for (int j = 0; j < 2; j++) {  
        for (int k = 0; k < 2; k++) {  
            arr[i][j][k] = sc.nextInt();  
        }  
    }  
}
```

```
// Assume first element is maximum
```

```
int max = arr[0][0][0];
```

```
// Find maximum element
```

```
for (int i = 0; i < 2; i++) {  
    for (int j = 0; j < 2; j++) {  
        for (int k = 0; k < 2; k++) {  
            if (arr[i][j][k] > max) {  
                max = arr[i][j][k];  
            }  
        }  
    }  
}
```

```
}
```

```
System.out.println("Maximum element = " + max);
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter 8 elements:

10

25

7

40

15

30

5

20

Maximum element = 40

18. Write a Java program to search for an element in a 3-D array and display its position.

```
import java.util.Scanner;
```

```
public class Search3DArray {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int[][][] arr = new int[2][2][2];
```

```
// Input elements
```

```
System.out.println("Enter 8 elements:");
```

```
for (int i = 0; i < 2; i++) {  
    for (int j = 0; j < 2; j++) {  
        for (int k = 0; k < 2; k++) {  
            arr[i][j][k] = sc.nextInt();  
        }  
    }  
}
```

```
// Input element to search
```

```
System.out.print("Enter element to search: ");
```

```
int search = sc.nextInt();
```

```
boolean found = false;
```

```
// Search element
```

```
for (int i = 0; i < 2; i++) {  
    for (int j = 0; j < 2; j++) {  
        for (int k = 0; k < 2; k++) {  
            if (arr[i][j][k] == search) {  
                System.out.println("Element found at position: ["  
                    + i + "][" + j + "][" + k + "]");  
                found = true;  
            }  
        }  
    }  
}
```

```

        }
    }
}

if (!found) {
    System.out.println("Element not found.");
}

sc.close();
}
}

```

OUTPUT:-

Enter 8 elements:

10

20

30

40

50

60

70

80

Enter element to search: 60

Element found at position: [1][0][1]

19. Write a Java program to create and display a jagged array with 3 rows of different lengths.

```
import java.util.Scanner;

public class JaggedArray {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Create jagged array with 3 rows
        int[][] arr = new int[3][];

        // Different lengths for each row
        arr[0] = new int[2];
        arr[1] = new int[3];
        arr[2] = new int[4];

        // Input elements
        System.out.println("Enter elements:");

        for (int i = 0; i < arr.length; i++) {

            System.out.println("Enter " + arr[i].length + " elements for row " + (i + 1) + ":");

            for (int j = 0; j < arr[i].length; j++) {

                arr[i][j] = sc.nextInt();

            }

        }

        // Display jagged array
```

```
System.out.println("Jagged Array:");
```

```
for (int i = 0; i < arr.length; i++) {  
    for (int j = 0; j < arr[i].length; j++) {  
        System.out.print(arr[i][j] + "\t");  
    }  
    System.out.println();  
}
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter elements:

Enter 2 elements for row 1:

10

20

Enter 3 elements for row 2:

30

40

50

Enter 4 elements for row 3:

60

70

80

90

Jagged Array:

```
10    20
30    40    50
60    70    80    90
```

20. Create a Java program to store marks for students where each student has taken a different number of subjects, then display them.

```
import java.util.Scanner;
```

```
public class StudentMarks {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Jagged array: different subjects for each student
        int[][] marks = new int[3][];

        // Different number of subjects
        marks[0] = new int[3];
        marks[1] = new int[4];
        marks[2] = new int[2];

        // Input marks
        for (int i = 0; i < marks.length; i++) {
            System.out.println("Enter marks for Student " + (i + 1) + " :");

            for (int j = 0; j < marks[i].length; j++) {
                System.out.print("Subject " + (j + 1) + " : ");
            }
        }
    }
}
```

```
        marks[i][j] = sc.nextInt();
    }
}

// Display marks

System.out.println("\nStudent Marks:");

for (int i = 0; i < marks.length; i++) {
    System.out.print("Student " + (i + 1) + ": ");

    for (int j = 0; j < marks[i].length; j++) {
        System.out.print(marks[i][j] + " ");
    }

    System.out.println();
}

sc.close();
}
```

OUTPUT:-

Enter marks for Student 1:

Subject 1: 80

Subject 2: 75

Subject 3: 90

Enter marks for Student 2:

Subject 1: 85

Subject 2: 70

Subject 3: 88

Subject 4: 92

Enter marks for Student 3:

Subject 1: 78

Subject 2: 84

Student Marks:

Student 1: 80 75 90

Student 2: 85 70 88 92

Student 3: 78 84

21. Write a program to rotate an array to the left by one position.

```
import java.util.Scanner;
```

```
public class LeftRotate {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int[] arr = new int[5];  
  
        // Input array elements  
        System.out.println("Enter 5 integers:");  
        for (int i = 0; i < 5; i++) {  
            arr[i] = sc.nextInt();  
        }  
    }  
}
```

```
// Store the first element
int first = arr[0];

// Shift elements to the left
for (int i = 0; i < 4; i++) {
    arr[i] = arr[i + 1];
}

// Put first element at the end
arr[4] = first;

// Display rotated array
System.out.println("Array after left rotation:");
for (int i = 0; i < 5; i++) {
    System.out.print(arr[i] + " ");
}

sc.close();
}
```

OUTPUT:-

Enter 5 integers:

10

20

30

40

50

Array after left rotation:

20 30 40 50 10

22. Count how many times each element appears in a 1-D array.

```
import java.util.Scanner;
```

```
public class ElementFrequency {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int[] arr = new int[5];  
        boolean[] visited = new boolean[5];  
  
        // Input array elements  
        System.out.println("Enter 5 integers:");  
        for (int i = 0; i < 5; i++) {  
            arr[i] = sc.nextInt();  
        }  
  
        // Count frequency of each element  
        System.out.println("Frequency of each element:");  
  
        for (int i = 0; i < 5; i++) {  
  
            if (visited[i]) {
```

```
        continue;
    }
}
```

```
int count = 1;
```

```
for (int j = i + 1; j < 5; j++) {
    if (arr[i] == arr[j]) {
        count++;
        visited[j] = true;
    }
}
```

```
System.out.println(arr[i] + " appears " + count + " time(s)");
}
```

```
sc.close();
}
}
```

OUTPUT:-

Enter 5 integers:

10

20

10

30

20

Frequency of each element:

10 appears 2 time(s)

20 appears 2 time(s)

30 appears 1 time(s)

23. Find the second largest element in a 1-D array.

```
import java.util.Scanner;
```

```
public class SecondLargest {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int[] arr = new int[5];  
  
        // Input array elements  
        System.out.println("Enter 5 integers:");  
        for (int i = 0; i < 5; i++) {  
            arr[i] = sc.nextInt();  
        }  
  
        int largest = Integer.MIN_VALUE;  
        int secondLargest = Integer.MIN_VALUE;  
  
        // Find largest and second largest  
        for (int i = 0; i < 5; i++) {  
            if (arr[i] > largest) {  
                secondLargest = largest;  
                largest = arr[i];  
            }  
        }  
    }  
}
```

```

        largest = arr[i];
    } else if (arr[i] > secondLargest && arr[i] != largest) {
        secondLargest = arr[i];
    }
}

if (secondLargest == Integer.MIN_VALUE) {
    System.out.println("Second largest element does not exist.");
} else {
    System.out.println("Second largest element = " + secondLargest);
}

sc.close();
}
}

```

OUTPUT:-

Enter 5 integers:

10

45

25

60

30

Second largest element = 45

24. Remove duplicate elements from a 1-D array (without using collections).

```
import java.util.Scanner;
```

```
public class RemoveDuplicates {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int[] arr = new int[5];  
        int[] unique = new int[5];  
        int count = 0;  
  
        // Input array elements  
        System.out.println("Enter 5 integers:");  
        for (int i = 0; i < 5; i++) {  
            arr[i] = sc.nextInt();  
        }  
  
        // Remove duplicate elements  
        for (int i = 0; i < 5; i++) {  
            boolean duplicate = false;  
  
            for (int j = 0; j < count; j++) {  
                if (arr[i] == unique[j]) {  
                    duplicate = true;  
                    break;  
                }  
            }  
  
            if (!duplicate) {
```

```

        unique[count] = arr[i];
        count++;
    }
}

// Display array without duplicates
System.out.println("Array after removing duplicates:");

for (int i = 0; i < count; i++) {
    System.out.print(unique[i] + " ");
}

sc.close();
}
}

```

OUTPUT:-

Enter 5 integers:

10

20

10

30

20

Array after removing duplicates:

10 20 30

25. Find the sum of the main diagonal and secondary diagonal elements of a square matrix.

```
import java.util.Scanner;
```

```
public class DiagonalSum {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int[][] matrix = new int[3][3];  
        int mainSum = 0;  
        int secondarySum = 0;  
  
        // Input matrix elements  
        System.out.println("Enter 9 elements of the matrix:");  
  
        for (int i = 0; i < 3; i++) {  
            for (int j = 0; j < 3; j++) {  
                matrix[i][j] = sc.nextInt();  
            }  
        }  
  
        // Calculate diagonal sums  
        for (int i = 0; i < 3; i++) {  
            mainSum = mainSum + matrix[i][i];  
            secondarySum = secondarySum + matrix[i][2 - i];  
        }  
  
        // Display matrix  
        System.out.println("Matrix:");
```

```
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j < 3; j++) {  
        System.out.print(matrix[i][j] + "\t");  
    }  
    System.out.println();  
}
```

```
System.out.println("Sum of main diagonal = " + mainSum);
```

```
System.out.println("Sum of secondary diagonal = " + secondarySum);
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter 9 elements of the matrix:

1

2

3

4

5

6

7

8

9

Matrix:

1	2	3
4	5	6
7	8	9

Sum of main diagonal = 15

Sum of secondary diagonal = 15

26. Display the upper and lower triangular parts of a square matrix separately.

```
import java.util.Scanner;
```

```
public class TriangularMatrix {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int[][] matrix = new int[3][3];
```

```
        // Input matrix elements
```

```
        System.out.println("Enter 9 elements of the matrix:");
```

```
        for (int i = 0; i < 3; i++) {
```

```
            for (int j = 0; j < 3; j++) {
```

```
                matrix[i][j] = sc.nextInt();
```

```
            }
```

```
        }
```

```
        // Display Upper Triangular Matrix
```

```
        System.out.println("Upper Triangular Matrix:");
```

```
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j < 3; j++) {  
        if (i <= j) {  
            System.out.print(matrix[i][j] + "\t");  
        } else {  
            System.out.print("0\t");  
        }  
    }  
    System.out.println();  
}
```

// Display Lower Triangular Matrix

```
System.out.println("Lower Triangular Matrix:");
```

```
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j < 3; j++) {  
        if (i >= j) {  
            System.out.print(matrix[i][j] + "\t");  
        } else {  
            System.out.print("0\t");  
        }  
    }  
    System.out.println();  
}
```

```
        sc.close();
    }
}
```

OUTPUT:-

Enter 9 elements of the matrix:

1

2

3

4

5

6

7

8

9

Upper Triangular Matrix:

1	2	3
---	---	---

0	5	6
---	---	---

0	0	9
---	---	---

Lower Triangular Matrix:

1	0	0
---	---	---

4	5	0
---	---	---

7	8	9
---	---	---

27. Print only the boundary elements of a matrix.

```
import java.util.Scanner;
```

```
public class BoundaryElements {
```

```
public static void main(String[] args) {  
    Scanner sc = new Scanner(System.in);  
  
    int[][] matrix = new int[3][3];  
  
    // Input matrix elements  
    System.out.println("Enter 9 elements of the matrix:");  
  
    for (int i = 0; i < 3; i++) {  
        for (int j = 0; j < 3; j++) {  
            matrix[i][j] = sc.nextInt();  
        }  
    }  
  
    // Display only boundary elements  
    System.out.println("Boundary elements:");  
  
    for (int i = 0; i < 3; i++) {  
        for (int j = 0; j < 3; j++) {  
            if (i == 0 || i == 2 || j == 0 || j == 2) {  
                System.out.print(matrix[i][j] + "\t");  
            } else {  
                System.out.print(" \t");  
            }  
        }  
    }  
    System.out.println();  
}
```

```
}
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT:-

Enter 9 elements of the matrix:

1

2

3

4

5

6

7

8

9

Boundary elements:

1 2 3

4 6

7 8 9

28. Search for an element in a 2-D array and display its row & column position.

```
import java.util.Scanner;
```

```
public class Search2DArray {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
int[][] matrix = new int[3][3];
```

```
// Input matrix elements
```

```
System.out.println("Enter 9 elements of the matrix:");
```

```
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j < 3; j++) {  
        matrix[i][j] = sc.nextInt();  
    }  
}
```

```
// Input element to search
```

```
System.out.print("Enter element to search: ");
```

```
int search = sc.nextInt();
```

```
boolean found = false;
```

```
// Search the element
```

```
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j < 3; j++) {  
        if (matrix[i][j] == search) {  
            System.out.println("Element found at row "  
                + (i + 1) + ", column " + (j + 1));  
            found = true;  
        }  
    }  
}
```

```
    }  
}
```

```
if (!found) {  
    System.out.println("Element not found.");  
}
```

```
    sc.close();  
}  
}
```

OUTPUT:-

Enter 9 elements of the matrix:

10

20

30

40

50

60

70

80

90

Enter element to search: 60

Element found at row 2, column 3

29. Sort the elements of each row in a jagged array.

```
import java.util.Scanner;
```

```
public class SortJaggedArray {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        // Create jagged array with 3 rows  
        int[][] arr = new int[3][];  
  
        // Different lengths for each row  
        arr[0] = new int[3];  
        arr[1] = new int[4];  
        arr[2] = new int[2];  
  
        // Input elements  
        System.out.println("Enter elements:");  
  
        for (int i = 0; i < arr.length; i++) {  
            System.out.println("Enter " + arr[i].length +  
                " elements for row " + (i + 1) + ":");  
  
            for (int j = 0; j < arr[i].length; j++) {  
                arr[i][j] = sc.nextInt();  
            }  
        }  
  
        // Sort each row using bubble sort  
        for (int i = 0; i < arr.length; i++) {
```

```

        for (int j = 0; j < arr[i].length - 1; j++) {
            for (int k = 0; k < arr[i].length - j - 1; k++) {
                if (arr[i][k] > arr[i][k + 1]) {
                    int temp = arr[i][k];
                    arr[i][k] = arr[i][k + 1];
                    arr[i][k + 1] = temp;
                }
            }
        }
    }
}

// Display sorted jagged array
System.out.println("\nJagged Array after sorting each row:");

for (int i = 0; i < arr.length; i++) {
    for (int j = 0; j < arr[i].length; j++) {
        System.out.print(arr[i][j] + "\t");
    }
    System.out.println();
}

sc.close();
}
}

```

OUTPUT:-

Enter elements:

Enter 3 elements for row 1:

30

10

20

Enter 4 elements for row 2:

80

40

60

50

Enter 2 elements for row 3:

90

70

Jagged Array after sorting each row:

10 20 30

40 50 60 80

70 90

30. Check if a given matrix is a sparse matrix (more zero elements than non-zero).

```
import java.util.Scanner;
```

```
public class SparseMatrix {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int[][] matrix = new int[3][3];
```

```
        int zeroCount = 0;
```

```
        int nonZeroCount = 0;
```

```
// Input matrix elements
```

```
System.out.println("Enter 9 elements of the matrix:");
```

```
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j < 3; j++) {  
        matrix[i][j] = sc.nextInt();  
  
        if (matrix[i][j] == 0) {  
            zeroCount++;  
        } else {  
            nonZeroCount++;  
        }  
    }  
}
```

```
// Display matrix
```

```
System.out.println("Matrix:");
```

```
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j < 3; j++) {  
        System.out.print(matrix[i][j] + " ");  
    }  
    System.out.println();  
}
```

```
// Check sparse matrix
if (zeroCount > nonZeroCount) {
    System.out.println("The matrix is a Sparse Matrix.");
} else {
    System.out.println("The matrix is not a Sparse Matrix.");
}

System.out.println("Number of zero elements = " + zeroCount);
System.out.println("Number of non-zero elements = " + nonZeroCount);

sc.close();
}
}
```

OUTPUT:-

Enter 9 elements of the matrix:

0

0

5

0

2

0

0

0

3

Matrix:

0	0	5
0	2	0
0	0	3

The matrix is a Sparse Matrix.

Number of zero elements = 6

Number of non-zero elements = 3